

Optimization

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Overview

Course info

Vector Space and Euclidean Space

Convex Sets and Some Examples

Operations that preserve convexity

Generalized inequalities

Separating & supporting hyperplanes

Summary

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Topics in This Course

- ▶ Widely used computational problem solving methodology.
- ▶ Fundamental techniques in AI and computer science.

- ▶ Traditional Opt (8 weeks)
- ▶ Opt Algorithms for Machine Learning (1-2 weeks)
- ▶ Modern heuristics and Multiobjective Opt (1-2 weeks)

Pre-requisites: basic calculus and linear algebra.

Ref Books and Resources

- ▶ Convex Optimization, Stephen Boyd and Lieven Vandenberghe, Cambridge University Press.
- ▶ Convex Optimization: Algorithms and Complexity, Sébastien Bubeck.
- ▶ Convex Analysis and Nonlinear Optimization, Theory and Examples, Jonathan Borwein , Adrian Lewis.
- ▶ Handbook of Metaheuristics (International Series in Operations Research & Management Science) 2nd ed. 2010, Michel Gendreau (Editor), Jean-Yves Potvin (Editor).
- ▶ Papers et al.

Acknowledgment: Thanks to Prof. S Boyd for course materials in this course. Most materials from week 1 to 8 are from his slides for convex opt.

Assessment

- ▶ Course work: 40%
 - ▶ Tutorial exercises and take-home assignments 20%
 - ▶ Midterm 20%
- ▶ Examination: 60%

How to study

- ▶ Read notes before lecture and take notes during lecture.
- ▶ Be active. Ask questions if you have.
- ▶ Do some mini-research on optimization.

An optimization problem can be written as

$$\begin{aligned} \min \quad & f(x) \\ \text{s.t.} \quad & x \in \Omega \end{aligned} \tag{1}$$

where $f : \Omega \rightarrow R$ is the objective function.

- ▶ globally optimal solution x^* : $f(x^*) \leq f(x)$ for any $x \in \Omega$
- ▶ locally optimal solution x^* : there exists $N(x^*)$, a neighborhood of x^* , such that $f(x^*) \leq f(x)$ for any $x \in \Omega \cap N(x^*)$.
- ▶ Discrete opt, continuous opt. multi-objective opt

An example:

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Vector Space

A vector space is a nonempty set V of objects (called vectors) with two operations: addition (+) and multiplication by real numbers (*), subject to the following properties for any $u, v, w \in V$ and $a, b \in R$:

1. $u + v \in V, a * u \in V$.
2. $u + v = v + u$
3. $u + (v + w) = (u + v) + w$
4. There exists a zero vector $\mathbf{0}$ such that $u + \mathbf{0} = u$.
5. For each $u \in V$, there exists $-u \in V$ such that $u + (-u) = \mathbf{0}$
6. $a(u + v) = au + av, (a + b)u = au + bu$
7. $1u = u,$
8. $(ab)u = a(bu)$

Q: Is zero vector unique? Why

- ▶ Finite dimensional vector space
- ▶ Examples: R^n (n -D real vector space), S^n

Finite-Dimensional Euclidean Space

Let V be a vector space, an inner product $\langle \cdot, \cdot \rangle$ is a mapping:
 $V \times V \rightarrow \mathbb{R}$ such that, for any $u, v, w \in V$ and $a \in \mathbb{R}$

- ▶ $\langle v, v \rangle \geq 0$; " $=$ " iff $v = 0$,
- ▶ $\langle v, u \rangle = \langle u, v \rangle$.
- ▶ $\langle av + bu, w \rangle = a\langle v, w \rangle + b\langle u, w \rangle$.

Norm induced by the inner product:

$$\|v\| = \sqrt{\langle v, v \rangle}$$

We can prove:

$$|\langle v, u \rangle| \leq \|v\| * \|u\|$$

Finite-dimensional Euclidean space E : a finite-dimensional vector space with an inner product $\langle \cdot, \cdot \rangle$.

Two Examples:

- ▶ $\mathbb{R}^n = ?$ $\langle x, y \rangle = x^T y = \sum x_i y_i$.
- ▶ $S^n =$ the set of all the $n \times n$ symmetric matrices.
 $\langle A, B \rangle = \text{tr}(AB)$.

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Convex sets

Line segment between x_1 and x_2 : the set of all points of form

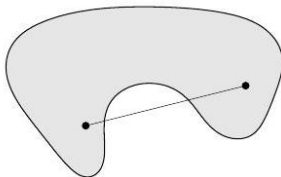
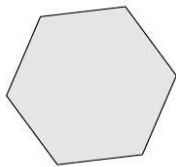
$$x = \theta x_1 + (1 - \theta)x_2$$

with $0 \leq \theta \leq 1$.

Convex set: contains line segment between any two points in the set

$$x_1, x_2 \in C, \quad 0 \leq \theta \leq 1 \quad \implies \quad \theta x_1 + (1 - \theta)x_2 \in C$$

examples: (one convex, two nonconvex sets)



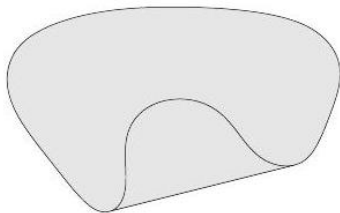
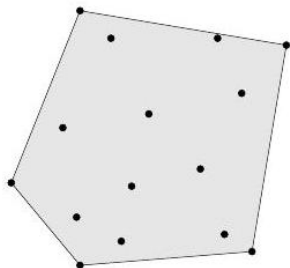
Convex combination and convex hull

Convex combination of $x_1, \dots, x_k$: any point x of the form

$$x = \theta_1 x_1 + \theta_2 x_2 + \dots + \theta_k x_k$$

with $\theta_1 + \dots + \theta_k = 1, \theta_i \geq 0$.

Convex hull: $\text{conv}(S)$ = set of all convex combinations of points in S

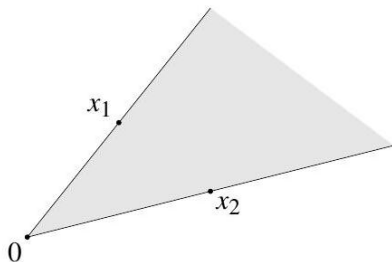


Convex cone

conic (nonnegative) combination of x_1 and x_2 : any point of the form

$$x = \theta_1 x_1 + \theta_2 x_2$$

with $\theta_1 \geq 0, \theta_2 \geq 0$

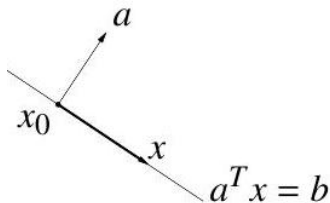


Convex cone: set that contains all conic combinations of points in the set

Hyperplanes and halfspaces

hyperplane: set of the form $\{x \mid a^T x = b\}$, with $a \neq 0$.

halfspace: set of the form $\{x \mid a^T x \leq b\}$, with $a \neq 0$



- ▶ a is the normal vector
- ▶ hyperplanes are convex; halfspaces are convex

Euclidean balls and ellipsoids

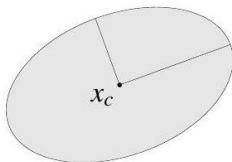
(Euclidean) ball with center x_c and radius r :

$$B(x_c, r) = \{x \mid \|x - x_c\|_2 \leq r\} = \{x_c + ru \mid \|u\|_2 \leq 1\}$$

ellipsoid: set of the form

$$\{x \mid (x - x_c)^T P^{-1} (x - x_c) \leq 1\}$$

with $P \in \mathbf{S}_{++}^n$ (i.e., P symmetric positive definite)



another representation: $\{x_c + Au \mid \|u\|_2 \leq 1\}$ with A square and nonsingular

Norm balls and norm cones

norm: a function $\|\cdot\|$ that satisfies:

- ▶ $\|x\| \geq 0$; $\|x\| = 0$ if and only if $x = 0$
- ▶ $\|tx\| = |t|\|x\|$ for $t \in \mathbf{R}$
- ▶ $\|x + y\| \leq \|x\| + \|y\|$

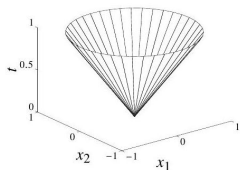
notation: $\|\cdot\|$ is general (unspecified) norm; $\|\cdot\|_{\text{symb}}$ is particular norm

- ▶ norm ball with center x_c and radius r : $\{x \mid \|x - x_c\| \leq r\}$
- ▶ norm cone: $\{(x, t) \mid \|x\| \leq t\}$
- ▶ norm balls and cones are convex

Euclidean norm cone

$$\{(x, t) \mid \|x\|_2 \leq t\} \subset \mathbf{R}^{n+1}$$

is called second-order cone



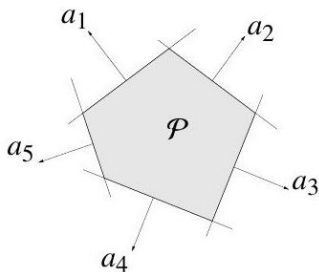
Polyhedra

- ▶ polyhedron is solution set of finitely many linear inequalities and equalities

$$\{x \mid Ax \leq b, Cx = d\}$$

($A \in \mathbf{R}^{m \times n}$, $C \in \mathbf{R}^{p \times n}$, $\leq$ is componentwise inequality)

- ▶ intersection of finite number of halfspaces and hyperplanes
- ▶ example with no equality constraints; a_i^T are rows of A



Positive semidefinite cone

- ▶ $\mathbf{S}^n$ is set of symmetric $n \times n$ matrices
- ▶ $\mathbf{S}_+^n = \{X \in \mathbf{S}^n \mid X \geq 0\}$: positive semidefinite (symmetric) $n \times n$ matrices. $X \in \mathbf{S}_+^n \iff z^T X z \geq 0$ for all z
- ▶ $\mathbf{S}_+^n$ is a convex cone, the positive semidefinite cone
- ▶ $\mathbf{S}_{++}^n = \{X \in \mathbf{S}^n \mid X > 0\}$: positive definite (symmetric) $n \times n$ matrices

Example:

$$\begin{bmatrix} x & y \\ y & z \end{bmatrix} \in \mathbf{S}_+^2$$

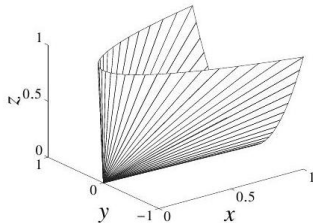


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Showing a set is convex

methods for establishing convexity of a set C

1. apply definition: show

$$x_1, x_2 \in C, 0 \leq \theta \leq 1 \implies \theta x_1 + (1 - \theta)x_2 \in C$$

▶ recommended only for very simple sets

2. use convex functions (next lecture)

3. show that C is obtained from simple convex sets (hyperplanes, halfspaces, norm balls, ...) by operations that preserve convexity

▶ intersection

▶ affine mapping

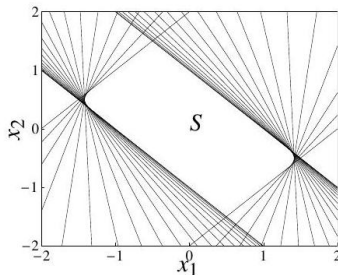
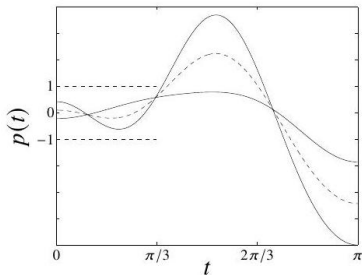
▶ perspective mapping

▶ linear-fractional mapping

you'll mostly use methods 2 and 3

Intersection

- ▶ the intersection of (any number of) convex sets is convex
- ▶ example:
 - ▶ $S = \{x \in \mathbf{R}^m \mid |p(t)| \leq 1 \text{ for } |t| \leq \pi/3\}$ with
 $p(t) = x_1 \cos t + \cdots + x_m \cos mt$
 - ▶ write $S = \bigcap_{|t| \leq \pi/3} \{x \mid |p(t)| \leq 1\}$, i.e., an intersection of (convex) slabs
- ▶ picture for $m = 2$:



Affine mappings

suppose $f : \mathbf{R}^n \rightarrow \mathbf{R}^m$ is affine, i.e., $f(x) = Ax + b$ with $A \in \mathbf{R}^{m \times n}, b \in \mathbf{R}^m$

- ▶ the image of a convex set under f is convex

$$S \subseteq \mathbf{R}^n \text{ convex} \implies f(S) = \{f(x) \mid x \in S\} \text{ convex}$$

- ▶ the inverse image $f^{-1}(C)$ of a convex set under f is convex

$$C \subseteq \mathbf{R}^m \text{ convex} \implies f^{-1}(C) = \{x \in \mathbf{R}^n \mid f(x) \in C\} \text{ convex}$$

Examples

- ▶ scaling, translation: $aS + b = \{ax + b \mid x \in S\}$, $a, b \in \mathbf{R}$
- ▶ projection onto some coordinates: $\{x \mid (x, y) \in S\}$
- ▶ if $S \subseteq \mathbf{R}^n$ is convex and $c \in \mathbf{R}^n$, $c^T S = \{c^T x \mid x \in S\}$ is an interval
- ▶ solution set of linear matrix inequality
 $\{x \mid x_1 A_1 + \cdots + x_m A_m \leq B\}$ with $A_i, B \in \mathbf{S}^p$
- ▶ hyperbolic cone $\{x \mid x^T P x \leq (c^T x)^2, c^T x \geq 0\}$ with
 $P \in \mathbf{S}_+^n$

Perspective and linear-fractional function

perspective function $P : \mathbf{R}^{n+1} \rightarrow \mathbf{R}^n$:

$$P(x, t) = x/t, \quad \text{dom } P = \{(x, t) \mid t > 0\}$$

(it is better to write it as ?)

- ▶ images and inverse images of convex sets under perspective are convex
- ▶ linear-fractional function $f : \mathbf{R}^n \rightarrow \mathbf{R}^m$:

$$f(x) = \frac{Ax + b}{c^T x + d}, \quad \text{dom } f = \{x \mid c^T x + d > 0\}$$

- ▶ images and inverse images of convex sets under linear-fractional functions are convex

Linear-fractional function example

$$f(x) = \frac{1}{x_1 + x_2 + 1} x$$

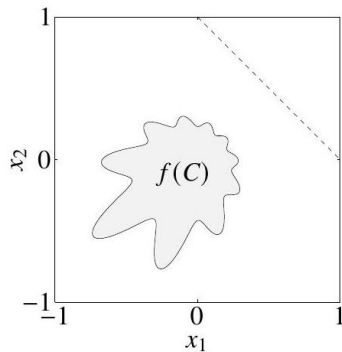
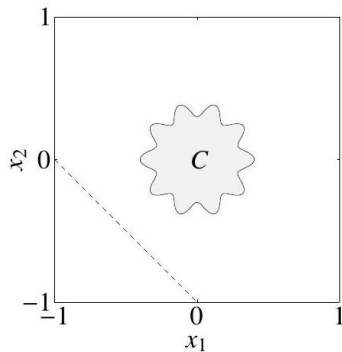


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Proper cones

a convex cone $K \subseteq \mathbf{R}^n$ is a proper cone if

- ▶ K is closed (contains its boundary)
- ▶ K is solid (has nonempty interior)
- ▶ K is pointed (contains no line)

examples

- ▶ nonnegative orthant
 $K = \mathbf{R}_+^n = \{x \in \mathbf{R}^n \mid x_i \geq 0, i = 1, \dots, n\}$
- ▶ positive semidefinite cone $K = \mathbf{S}_+^n$
- ▶ nonnegative polynomials on $[0, 1]$:

$$K = \{x \in \mathbf{R}^n \mid x_1 + x_2 t + x_3 t^2 + \dots + x_n t^{n-1} \geq 0 \text{ for } t \in [0, 1]\}$$

Generalized inequality

- ▶ (nonstrict and strict) generalized inequality defined by a proper cone K :

$$x \leq_K y \iff y - x \in K, \quad x <_K y \iff y - x \in \text{int } K$$

- ▶ examples

- ▶ componentwise inequality

$$(K = \mathbf{R}_+^n) : x \leq_{\mathbf{R}_+^n} y \iff x_i \leq y_i, \quad i = 1, \dots, n$$

- ▶ matrix inequality ($K = \mathbf{S}_+^n$) : $X \leq_{\mathbf{S}_+^n} Y \iff Y - X$ positive semidefinite

these two types are so common that we drop the subscript in $\leq_K$

- ▶ many properties of $\leq_K$ are similar to $\leq$ on $\mathbf{R}$, e.g.,

$$x \leq_K y, \quad u \leq_K v \implies x + u \leq_K y + v$$

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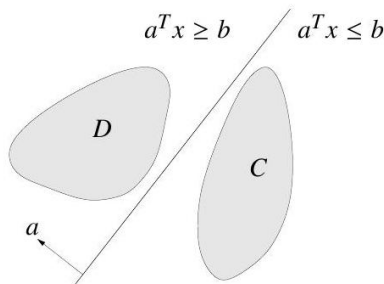
Separating & supporting hyperplanes

Summary

Separating hyperplane theorem

Theorem: if C and D are nonempty disjoint (i.e., $C \cap D = \emptyset$) convex sets, there exist $a \neq 0$, b s.t.

$$a^T x \leq b \text{ for } x \in C, \quad a^T x \geq b \text{ for } x \in D$$



- ▶ the hyperplane $\{x \mid a^T x = b\}$ separates C and D
- ▶ strict separation (" $\leq$ " and " $\geq$ " are replaced by " $<$ " and " $>$ ", respectively.) requires additional assumptions (e.g., C is closed, D is a singleton)

Outline of Proof

Case 1: D has only one element d and $\text{dist}(C, d) > 0$ (i.e. $d \notin \text{Bd}(C)$)

- ▶ Let $\text{cl}(C)$ be the smallest closed set including C , i.e.

$$\text{cl}(C) = C \cup \text{bd}(C)$$

- ▶ $\text{cl}(C)$ is convex, $\text{dist}(\text{cl}(C), d) = \text{dist}(C, d)$, and $C \subset \text{cl}(C)$
- ▶ without loss of generality, we assume C is closed.

1. There exists a unique $c \in C$ such that $\text{dist}(c, d) = \text{dist}(C, d) > 0$ (why)
2. Let $a = d - c \neq 0$, $x_0 = \frac{d+c}{2}$. Consider $f(x) = \frac{a^T}{\|a\|}(x - x_0)$.
3. $f(d) < 0$
4. for any $x \in C$, $f(x) > 0$ (How to prove? by contradiction)

strict separation!

Outline of Proof

Case 2: D has only one element d and $\text{dist}(C, d) = 0$ (i.e. $d \in \text{bd}(C)$.)

- ▶ For any integer m , there exists $d_m \notin \text{cl}(C)$ and $\text{dist}(d_m, d) < \frac{1}{m}$.
- ▶ By Case 1, there exists affine function $f_m(x) = \frac{a_m^T}{\|a_m\|}(x - x_m)$ such that

$$f_m(d_m) < 0, \text{ and } f_m(x) > 0 \text{ for all } x \in C.$$

- ▶ There exists a sub-sequence $m_1 < m_2 < \dots < m_k \dots$ such that

$$\frac{a_m^T}{\|a_m\|} \rightarrow a, \text{ and } x_m \rightarrow x_0$$

- ▶ Let $f(x) = a^T(x - x_0)$, then

$$f(d) \geq 0, \text{ and } f(x) \leq 0 \text{ for all } x \in C.$$

Outline of Proof

Case 3: $D \cap C = \emptyset$ (general case).

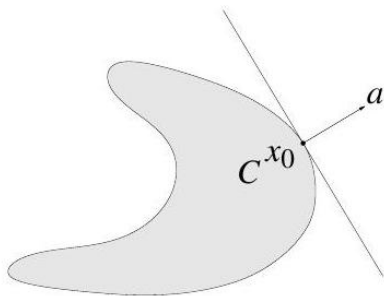
- ▶ Let $F = D - C = \{x - y | x \in D \text{ and } y \in C\}$ (F is convex)
- ▶ $0 \notin F$
- ▶ there exists $a^T x = b$ separating 0 and F . Without loss of generality, Assume

$$a^T 0 + b \geq 0, \text{ and } a^T(x - y) + b \leq 0 \text{ for any } x \in D, y \in C$$

- ▶ $a^T x \leq a^T y$ for any $x \in D, y \in C$
- ▶ Let $\alpha = \max\{a^T x | x \in D\}$, then $a^T x = \alpha$ is an separating plane for D and C .

Supporting hyperplane theorem

- ▶ suppose x_0 is a boundary point of set $C \subset \mathbf{R}^n$
- ▶ supporting hyperplane to C at x_0 has form $\{x \mid a^T x = a^T x_0\}$, where $a \neq 0$ and $a^T x \leq a^T x_0$ for all $x \in C$



Theorem: supporting hyperplane theorem: if C is convex, then there exists a supporting hyperplane at every boundary point of C
Pf: Noting in the proof of case 2, we only require that

$$d \in Bd(C).$$

Theorem of Alternative

Gordan Theorem: For any $a^1, \dots, a^m \in R^n$, exactly one of the following systems has a solution:

$$(1) : \sum_{i=1}^m \lambda_i a^i = 0, \sum_{i=1}^m \lambda_i = 1, 0 \leq \lambda_1, \dots, \lambda_m \in R$$

$$(2) : a^{iT} x < 0 \text{ for } i = 1, \dots, m, x \in R^n$$

Outline of Pf: Noting that

- ▶ system (1) has a solution $\Leftrightarrow 0 \in \text{convhull}(a^1, \dots, a^m)$.
- ▶ system (2) has a solution $\Leftrightarrow$ there is an separating plane btw 0 and $\text{convhull}(a^1, \dots, a^m)$.

Farkas Lemma

Theorem: For any $a^1, \dots, a^m$ and $c \in R^n$, exactly one of the following systems has a solution:

$$(1) : \sum_{i=1}^m \lambda_i a^i = c, 0 \leq \lambda_1, \dots, \lambda_m \in R$$

$$(2) : (a^i)^T x \leq 0 \text{ for } i = 1, \dots, m, c^T x > 0, x \in R^n$$

Outline of Pf: Noting that

- ▶ system (1) has a solution $\Leftrightarrow 0 \in \text{convexcone}(a^1, \dots, a^m)$.
- ▶ system (2) has a solution $\Leftrightarrow$ there is an separating plane btw c and $\text{convexcone}(a^1, \dots, a^m)$.

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What we learned today

- ▶ Finite-D Euclidean space: R^n and S^n .
- ▶ Convex set
- ▶ How to prove a set is convex
- ▶ Separating and supporting hyperplanes
- ▶ Separating plane and Farkas Lemma.